

Northwind Technologies – CRM Implementation

Business analysis case study. Northwind Technologies and all names, dates and estimates are fictitious.

1. Business problem	
Current situation	Sales Representatives keep customer contacts in separate spreadsheets and inboxes. This causes duplicate and out-of-date records, no clear view of the sales pipeline, and missed meetings and follow-ups. Sales Managers have no easy way to report on team performance.
Proposed solution	A central CRM system with contact management, sales opportunity tracking, reporting dashboards, and Outlook email and calendar integration with automated reminders.

2. Objectives and success measures (illustrative targets)	
Accurate contact data	One central contact list; duplicate contacts below 2% three months after go-live.
Pipeline visibility	All open sales opportunities tracked in the CRM by sales stage.
Faster reporting	Sales Managers can produce a weekly sales report in under 10 minutes.
Fewer missed follow-ups	Automatic reminders for all booked meetings and follow-up tasks that are due.

3. Scope	
In scope	Contact management (including import and export); sales opportunity tracking; reporting and analytics dashboards; Outlook email and calendar integration with automated reminders; performance, security, privacy and audit NFRs.
Out of scope (assumed)	Marketing campaign automation; customer self-service portal; native mobile app; integration with finance or invoicing systems; migration of historical email archives.

4. Delivery approach		
Method	Scrum, 2-week sprints. 8 sprints (80 working days) plus 1 day for project closure = 81 working days.	
Planned start date	12 Oct 2026	<i>Linked to the Gantt Chart (cell E4) – change the date there to move the whole</i>

5. Stakeholders and project team		
Name	Role	Responsibility
Emily Johnson	Project Sponsor (assumed)	Approves budget and scope; signs off project completion (milestone M5).
Michael Lee	Head of Sales / Business Owner (assumed)	Represents sales team needs; signs off project completion (milestone M5).
Sarah	Product Owner	Owns and prioritises the backlog; accepts stories; owns milestones M1–M4.
David	Scrum Master	Runs Scrum events and removes blockers; leads project closure.
Tom	Business Analyst	Elicits and refines requirements, stories and acceptance criteria; maintains traceability.
Rachel	UX/UI Designer	Designs pages and user flows.
Lisa	Frontend Developer	Builds user-facing pages.
Mark	Backend Developer	Builds database, APIs and integrations.
Alex	QA / Tester	Writes and runs test cases from acceptance criteria; performance testing.
Sales Representatives	End users	Manage contacts and opportunities day to day.
Sales Managers	End users	Use reports and dashboards to track team performance.

6. Key assumptions	
•	Northwind uses Microsoft Outlook for email and calendar.
•	The Scrum team works full time on this project.
•	Durations are in working days (Mon–Fri); NZ public holidays and the Christmas break are not included.
•	Priorities, story points and success targets are illustrative for this exercise.

7. How to read this workbook	
Requirements	Functional requirements (REQ) and non-functional requirements (NFR).
EPICS	Four epics (EP-01 to EP-04), one per feature, with high-level acceptance criteria.
User Stories	EP-01 Contact Management split into 11 user stories with acceptance criteria, priority and story points.
Roadmap	High-level 17-week plan by phase.
Gantt Chart	Detailed schedule with owners, dependencies and milestones (formula-driven dates).
Traceability Matrix	Links each requirement to epics, user stories, tasks and test cases.

Risk Log	Key delivery risks with rating, mitigation and owner.
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Feature	Req ID
Contact Management	REQ-01
	REQ-02
Sales Opportunity Tracking	REQ-03
	REQ-04
Reporting and Analytics	REQ-05
	REQ-06
Integration and Automation	REQ-07
	REQ-08
System Performance NFRs	NFR-01
	NFR-02
Security & Privacy NFRs	NFR-03
	NFR-04
	NFR-05

Requirements
Enable sales representatives to list, view, add, edit, and delete customer contacts. Contact details must include customer name, company name, email, phone, additional details.
Allow users to import and export contact lists.
Implement functionality for users to create and update sales opportunities.
Provide a visual sales pipeline to track opportunities through different stages of the sales cycle.
Generate reports on sales performance and customer interactions.
Provide dashboard analytics to visualize sales data.
Integrate the CRM with email and calendar systems.
Automate reminders and follow-up tasks for sales representatives.
The system should be able to handle at least 500 concurrent users without any degradation in performance.
Page load times for any CRM feature must not exceed 3 seconds under normal network conditions.
Role-based access: users can only edit and delete records they are allowed to. Sales Representatives can delete only contacts and opportunities they own; Sales Managers can delete any record and view team-wide reports.
Privacy: customer personal information must be handled in line with the NZ Privacy Act 2020. Data is encrypted in transit and at rest, access is limited to authorised users, and a contact's details can be exported or removed on request.
Audit trail: the system records who created, changed or deleted each contact and opportunity, and when. Audit records are kept for at least 2 years.

Feature	Epic ID	EPIC
Contact Management	EP-01	As a Sales Representative, I want to manage all customer contacts in one central place, so that customer information is accurate, up to date and free of duplicates. Acceptance criteria: EP-01-01: Sales Representatives can list, view, add, edit and delete customer contacts (customer name, company name, email, phone and additional details). EP-01-02: Users can import and export contact lists.
Sales Opportunity Tracking	EP-02	As a Sales Representative, I want to create and update sales opportunities and see them by sales stage, so that I can prioritise leads and close deals faster. Acceptance criteria: EP-02-01: Users can create a sales opportunity linked to a contact and update its stage, value and expected close date. EP-02-02: A dashboard shows all opportunities grouped by stage of the sales cycle (e.g. Lead, Qualified, Proposal, Won, Lost).
Reporting and Analytics	EP-03	As a Sales Manager, I want to view dashboard analytics and sales reports, so that I can understand team performance and customer interactions and make better decisions. Acceptance criteria: EP-03-01: Sales Managers can view dashboard analytics that show team performance and customer interactions. EP-03-02: Sales Managers can generate and download sales reports.

Integration and Automation	EP-04	As a Sales Representative, I want the CRM to connect with my Outlook email and calendar and send me automatic reminders, so that I never miss a meeting or a customer follow-up. Acceptance criteria: EP-04-01: The CRM syncs with Outlook email and calendar; meetings booked in the CRM appear in Outlook, and customer emails are logged against the contact. EP-04-02: The system sends automatic reminders for upcoming meetings and follow-up tasks that are due.
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ID	User Story Title
US-01	List Customer Contacts
US-02	View Contact Details
US-03	Add New Contact
US-04	Validate Contact Details

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US-05	Edit Contact

US-06	Delete Contact

US-07	Search and Filter Contacts

US-08	Import Contacts

US-09	Handle Import Errors

US-10	Export Contacts

US-11	Contact Module Performance

Notes: Priority uses MoSCoW (Must / Should / Could / Won't). Story po.

User Story Description	Relates to Epic	AC ID
As a Sales Representative, I want to see a list of all customer contacts, so that I can quickly find the customer I need to contact.	EP-01	US-01-01
		US-01-02
		US-01-03
		US-01-04
As a Sales Representative, I want to open a contact and see all of its details, so that I have the full customer information before I call or email them.	EP-01	US-02-01
		US-02-02
		US-02-03
As a Sales Representative, I want to add a new customer contact, so that every new customer is recorded in one central place.	EP-01	US-03-01
		US-03-02
		US-03-03
As a Sales Representative, I want the system to check contact details when I add or edit a contact, so that records stay accurate and duplicates are avoided.	EP-01	US-04-01
		US-04-02

		US-04-03
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As a Sales Representative, I want to edit a contact's details, so that customer information stays up to date.	EP-01	US-05-01
		US-05-02
		US-05-03
		US-05-04

As a Sales Representative, I want to delete a contact I no longer need, so that the contact list stays clean and relevant.	EP-01	US-06-01
		US-06-02
		US-06-03
		US-06-04

As a Sales Representative, I want to search and filter contacts, so that I can find a customer quickly in a large list.	EP-01	US-07-01
		US-07-02
		US-07-03

As a Sales Representative, I want to import contacts from a CSV or Excel file, so that I can move existing spreadsheet data into the CRM without retyping it.	EP-01	US-08-01
		US-08-02
		US-08-03

As a Sales Representative, I want the system to tell me which rows could not be imported and why, so that I can fix my file and avoid bad or duplicate data.	EP-01	US-09-01
		US-09-02
		US-09-03

As a Sales Representative, I want to export contacts to a CSV or Excel file, so that I can share them or use them in other tools.	EP-01	US-10-01
		US-10-02
		US-10-03

As a Sales Representative, I want contact pages to load quickly even when the whole team is using the CRM, so that I can serve customers without delays.	EP-01	US-11-01
		US-11-02

ints use the Fibonacci scale (1, 2, 3, 5, 8) and are illustrative estimates for this exercise, sized to match the Gantt chart durations. Acc

Acceptance Criteria	Priority (MoSCoW)	Story Points
The list shows customer name, company name, email and phone for each contact.	Must	5
The list shows 25 contacts per page, sorted A–Z by customer name by default; the user can sort by any column.		
If there are no contacts, the message "No contacts found. Add your first contact." is shown with an Add Contact button.		
If the list cannot load (system error), an error message with a Retry button is shown and no partial data is displayed.		
Clicking a contact in the list opens a detail page showing customer name, company name, email, phone and additional details.	Must	3
The detail page shows the date the contact was created and when (and by whom) it was last updated.		
If the contact no longer exists (e.g. deleted by another user), the message "This contact is no longer available" is shown and the user returns to the list.		
An Add Contact form has fields for customer name, company name, email, phone and additional details; customer name and email are mandatory.	Must	3
Given the Add Contact form has valid details, When the user clicks Save, Then the contact is stored, appears in the contact list, and the message "Contact added" is shown.		
When the user clicks Cancel, the form closes and nothing is saved.		
If a mandatory field is empty, the contact is not saved and the field is highlighted with the message "This field is required".	Must	3
If the email format is invalid or the phone number contains letters, an error message is shown and the contact is not saved.		

Given a contact with the same email already exists, When the user tries to save a contact with that email, Then the warning "A contact with this email already exists" is shown with a link to the existing contact, and no duplicate is created.		
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The user can click Edit on the contact detail page, change any field and save the changes.	Must	3
The same validation rules as US-04 apply when editing.		
After saving, the updated details show in the list and detail page, and the "last updated" date and user are refreshed.		
If saving fails (system error), the user's changes are kept on screen, an error message is shown and the user can try again.		

When the user clicks Delete, a confirmation message "Are you sure you want to delete this contact?" is shown.	Should	2
If the user confirms, the contact is removed from the list and the message "Contact deleted" is shown.		
If the user cancels, the contact is not deleted.		
Given the contact is linked to an open sales opportunity, When the user clicks Delete, Then a warning is shown and the user must confirm before the contact is deleted.		

The user can search by customer name, company name or email; results update after at least 2 characters are typed.	Should	3
The user can filter the list by company name.		
If no contacts match, the message "No matching contacts found" is shown.		

The user can download an import template and upload a .csv or .xlsx file up to 5 MB.	Should	5
Before importing, the system shows a preview of the data and lets the user match file columns to contact fields.		
After import, a summary shows how many contacts were imported, skipped and failed.		

Unsupported file types or files over 5 MB are rejected with a clear error message.	Should	3
Given the import file has rows with missing mandatory fields or an invalid email, When the user imports the file, Then those rows are skipped and listed in an error report the user can download.		
Rows whose email matches an existing contact are flagged as duplicates, and the user chooses to skip or update them.		

The user can export all contacts, or only the current search/filter results, to .csv or .xlsx.	Should	2
The exported file includes column headers and all contact fields (customer name, company name, email, phone, additional details).		
If there are no contacts to export, the Export button is disabled.		

Load testing shows the contact module supports at least 500 concurrent users with no drop in performance.	Must	3
The contact list, detail, add and edit pages load in 3 seconds or less under normal network conditions.		

Acceptance criteria use two styles: rule-based statements and Given / When / Then (see US-03-02, US-04-03, US-06-04, US-09-02).

ID	Phase/Key Milestone	Duration	Week_01	Week_02	Week_03
1	Contact Management: Sales representatives can manage customer contacts easily.	20 days			
2	Sales Opportunity Tracking: Users can track sales opportunities and update their status.	10 days			
3	Reporting and Analytics: Sales managers can gain insights into team performance and customer interactions.	30 days			
4	Integration and Automation: CRM integrated with Outlook email & calendar, sales staff receive automated reminders about meetings and tasks	20 days			
5	Project Closure	1 day			
	Total	81 days			

Week_04	Week_05	Week_06	Week_07	Week_08	Week_09	Week_10	Week_11	Week_12

Week_13	Week_14	Week_15	Week_16	Week_17

Calendar date of each day column →

12/10

	Project : Northwind Technologies CRM Implementation	Resource	Duration	Start	End	
Ref	Phase, User Story/Task	Role or person	Days	Date	Date	1
	Project start date (Monday) – edit cell E4 to move the whole plan →			#####		
1.0	Phase 1: Contact Management (Epic EP-01)	Scrum Team	20	12/10/26	06/11/26	
1.1	US-01 List Customer Contacts	Scrum Team	7	12/10/26	20/10/26	
TASK-01	Refine story & acceptance criteria with Product Owner	Tom (BA)	1	12/10/26	12/10/26	
TASK-02	"Contact list" page design (after TASK-01)	Rachel (UX/UI)	1	13/10/26	13/10/26	
TASK-03	Contacts database table & list API – paging, sorting (after TASK-01)	Mark (Backend)	2	13/10/26	14/10/26	
TASK-04	Write test cases from acceptance criteria (after TASK-01)	Alex (QA)	1	13/10/26	13/10/26	
TASK-05	Build contact list page – table, paging, empty & error states (after TASK-02, TAS	Lisa (Frontend)	2	15/10/26	16/10/26	
TASK-06	Test and fix bugs (after TASK-05)	Alex (QA), Lisa, Mark	1	19/10/26	19/10/26	
TASK-07	Story review & acceptance sign-off (after TASK-06)	Tom (BA), Sarah (PO)	1	20/10/26	20/10/26	
1.2	US-02 View Contact Details (after TASK-05)	Scrum Team	3	19/10/26	21/10/26	
1.3	US-03 Add New Contact (after US-01)	Scrum Team	3	21/10/26	23/10/26	
1.4	US-04 Validate Contact Details (starts 1 day after US-03)	Scrum Team	3	22/10/26	26/10/26	
1.5	US-05 Edit Contact (after US-03)	Scrum Team	3	26/10/26	28/10/26	
1.6	US-06 Delete Contact (starts 1 day after US-05)	Scrum Team	2	27/10/26	28/10/26	
1.7	US-07 Search and Filter Contacts (after US-06)	Scrum Team	2	29/10/26	30/10/26	
1.8	US-08 Import Contacts (after US-05)	Scrum Team	3	29/10/26	02/11/26	
1.9	US-09 Handle Import Errors (starts 2 days after US-08)	Scrum Team	2	02/11/26	03/11/26	
1.10	US-10 Export Contacts (after US-08)	Scrum Team	2	03/11/26	04/11/26	
1.11	US-11 Contact Module Performance testing (after US-09)	Alex (QA), Mark	2	04/11/26	05/11/26	
1.12	Regression testing, Sprint review & demo (after US-11)	Scrum Team	1	06/11/26	06/11/26	
M1	◆ Milestone: Contact Management module complete	Sarah (PO)	0	06/11/26	06/11/26	
2.0	Phase 2: Sales Opportunity Tracking (Epic EP-02)	Scrum Team	10	09/11/26	20/11/26	
M2	◆ Milestone: Sales Opportunity Tracking live	Sarah (PO)	0	20/11/26	20/11/26	
3.0	Phase 3: Reporting and Analytics (Epic EP-03)	Scrum Team	30	23/11/26	01/01/27	
M3	◆ Milestone: Analytics dashboard launch	Sarah (PO)	0	01/01/27	01/01/27	

4.0	Phase 4: Integration and Automation (Epic EP-04)	Scrum Team	20	04/01/27	29/01/27	
M4	◆ Milestone: Outlook integration & automated reminders live	Sarah (PO)	0	29/01/27	29/01/27	
5.0	Phase 5: Project Closure – handover & lessons learned	David (SM), Sarah (PO)	1	01/02/27	01/02/27	
M5	◆ Milestone: Project complete	Emily Johnson, Michael Lee	0	01/02/27	01/02/27	

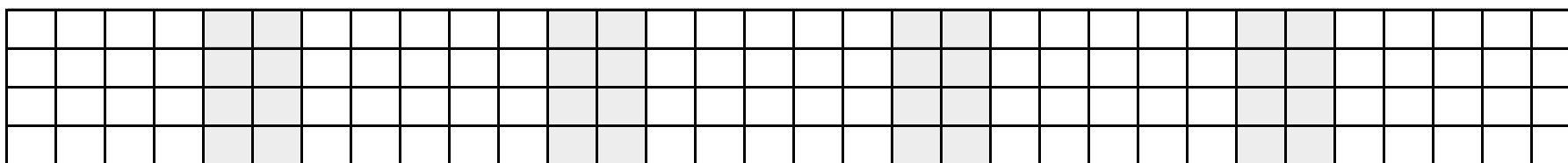
Legend:

	Phase
	User Story
	Task (US-01 List Customer Contacts)
	Milestone
	Weekend (no work planned)

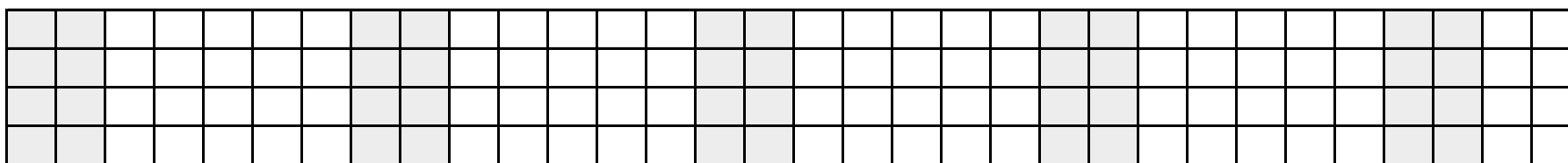
Assumptions:

- Durations are in working days (Mon–Fri). 1 sprint = 2 weeks = 10 working days. Each day column is one calendar day.
- Dependencies are written in brackets after each item and are built into the Start date formulas (WORKDAY).
- Public holidays (e.g. NZ Labour Day, Christmas/New Year) are not included – estimates are fictitious for this exercise.

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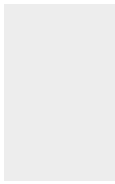
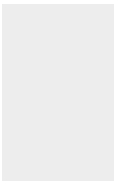


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Feature	Req ID	Requirements	User Story IDs
Contact Management	REQ-01	Enable sales representatives to list, view, add, edit, and delete customer contacts. Contact details must include customer name, company name, email, phone, additional details.	EP-01 (Epic) US-01, US-02, US-03, US-04, US-05, US-06, US-07
	REQ-02	Allow users to import and export contact lists.	EP-01 (Epic) US-08, US-09, US-10
Sales Opportunity Tracking	REQ-03	Implement functionality for users to create and update sales opportunities.	EP-02 (Epic)
	REQ-04	Provide a visual sales pipeline to track opportunities through different stages of the sales cycle.	EP-02 (Epic)
Reporting and Analytics	REQ-05	Generate reports on sales performance and customer interactions.	EP-03 (Epic)
	REQ-06	Provide dashboard analytics to visualize sales data.	EP-03 (Epic)
Integration and Automation	REQ-07	Integrate the CRM with email and calendar systems.	EP-04 (Epic)
	REQ-08	Automate reminders and follow-up tasks for sales representatives.	EP-04 (Epic)
System Performance	NFR-01	The system should be able to handle at least 500 concurrent users without any degradation in performance.	US-11 (applies to all features)
	NFR-02	Page load times for any CRM feature must not exceed 3 seconds under normal network conditions.	US-11 (applies to all features)

Security & Privacy	NFR-03	Role-based access: users can only edit and delete records they are allowed to. Sales Representatives can delete only contacts and opportunities they own; Sales Managers can delete any record and view team-wide reports.	Applies to all features; delete rules apply to US-06 Delete Contact
	NFR-04	Privacy: customer personal information must be handled in line with the NZ Privacy Act 2020. Data is encrypted in transit and at rest, access is limited to authorised users, and a contact's details can be exported or removed on request.	Applies to all features (US-01 to US-11, EP-02 to EP-04)
	NFR-05	Audit trail: the system records who created, changed or deleted each contact and opportunity, and when. Audit records are kept for at least 2 years.	US-02 View Contact Details, US-05 Edit Contact (partial)

Notes: each acceptance criterion is one test case (direct mapping). Test IDs match the acceptance criteria IDs on the EPICS and User Stories tabs. ID xx-yy (acceptance criterion / test case), TASK-xx (task).

Task IDs	Test IDs	Test Execution notes
TASK-01 to TASK-07 (US-01) Other stories: tasks defined at sprint planning	EP-01-01 US-01-01, US-01-02, US-01-03, US-01-04 US-02-01, US-02-02, US-02-03 US-03-01, US-03-02, US-03-03 US-04-01, US-04-02, US-04-03 US-05-01, US-05-02, US-05-03, US-05-04 US-06-01, US-06-02, US-06-03, US-06-04 US-07-01, US-07-02, US-07-03	Not yet executed. Planned for Sprint 1–2 testing; US-01 tests run with TASK-06.
Tasks defined at Sprint 2 planning	EP-01-02 US-08-01, US-08-02, US-08-03	Not yet executed. Planned for Sprint 2 testing.
Tasks defined at Sprint 3 planning	EP-02-01	Not yet executed. Epic to be split into user stories before Sprint 3.
Tasks defined at Sprint 3 planning	EP-02-02	Not yet executed. Epic to be split into user stories before Sprint 3.
Tasks defined at Sprint 4 planning	EP-03-02	Not yet executed. Epic to be split into user stories before Sprint 4.
Tasks defined at Sprint 4 planning	EP-03-01	Not yet executed. Epic to be split into user stories before Sprint 4.
Tasks defined at Sprint 7 planning	EP-04-01	Not yet executed. Epic to be split into user stories before Sprint 7.
Tasks defined at Sprint 7 planning	EP-04-02	Not yet executed. Epic to be split into user stories before Sprint 7.
Performance testing tasks (US-11)	US-11-01	Not yet executed. Load test in Sprint 2; repeat at end of each phase.
Performance testing tasks (US-11)	US-11-02	Not yet executed. Page load test in Sprint 2; repeat at end of each phase.

Defined at sprint planning	To be written: role permission tests (rep vs manager)	Not yet executed. Permission tests planned from Sprint 1 and repeated each phase.
Defined at sprint planning	To be written: privacy & security review checklist	Not yet executed. Privacy review planned before each phase goes live.
Defined at sprint planning	US-02-02, US-05-03 (partial); full audit tests to be written	Not yet executed. Partly tested in Sprint 1–2 with US-02 and US-05.

format used across the workbook: REQ-xx (functional requirement), NFR-xx (non-functional requirement), EP-xx (epic), US-xx (user story), US-

Risk ID	Risk description	Likelihood (1–3)	Impact (1–3)	Score	Rating	Mitigation	Owner	Status
R-01	No buffer at the end of Sprint 2: the 11 Contact Management stories only fit in 20 days if they run in parallel. Any delay pushes Phases 2–4 out.	3	2	6	High	Deliver Must stories first; move Should stories to a later sprint if needed; track the sprint burndown daily.	Sarah (PO)	Open
R-02	NZ public holidays and the Christmas/New Year break fall inside the plan but are not in the schedule, so the real end date will be later.	3	2	6	High	Add holidays to the WORKDAY calendar; agree a revised end date with the Project Sponsor.	David (SM)	Open
R-03	Outlook email and calendar integration is more complex than estimated (access permissions, sync conflicts).	2	3	6	High	Run a technical spike in Sprint 6, before Phase 4 starts; confirm Microsoft 365 access early.	Mark (Backend)	Open
R-04	Poor quality data in existing spreadsheets causes failed imports and duplicate contacts.	2	2	4	Medium	Import template, preview and error report (US-08, US-09); data clean-up with the sales team before go-live.	Tom (BA)	Open
R-05	Low user adoption: Sales Representatives keep using spreadsheets instead of the CRM.	2	3	6	High	Involve reps in sprint reviews; training at the end of each phase; managers report only from CRM data.	Michael Lee	Open
R-06	Breach of customer personal information.	1	3	3	Medium	Apply NFR-03 (role-based access) and NFR-04 (privacy); security review before each go-live.	Sarah (PO)	Open

Scoring: Score = Likelihood × Impact (1 = low, 3 = high). Rating: 6–9 High, 3–5 Medium, 1–2 Low. Blue numbers are inputs; Score and Rating are formulas.